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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 224 — Past Questions & Materials
by Renance

Course: EMT 224
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200

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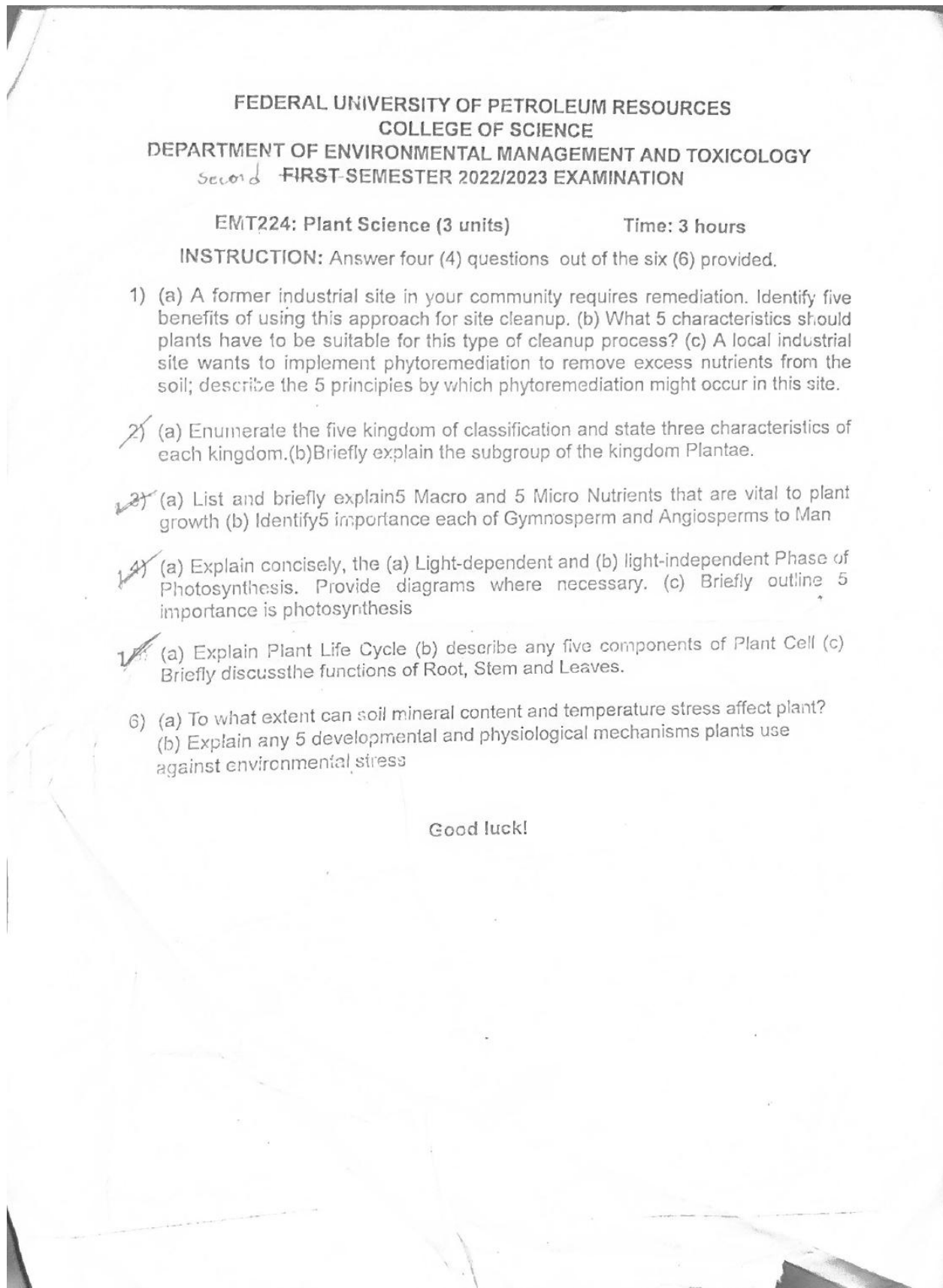
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Paper 1 of 5 — EMT 224 Plant Science 2022/2024

Level 200 • Second Semester • PastQuestion

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Paper 2 of 5 — EMT 224, 2023 past question

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF SCIENCE
DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY
SECOND SEMESTER 2022/2023 UNDERGRADUATE EXAMINATION

EMT224: Plant Science (3 units) Time: 3 hours

INSTRUCTION: Answer four (4) questions in all, one from each section

SECTION A

1. As organism, we perform three functions among others -- assimilation, reproduction and response to stimuli. In a unicellular organism, name five organelles, two of which are directly and three of which are indirectly associated with any two of these functions.
- x 2. No matter how tall a tree may be, water will always get to its apex from the root, not by active water transport, but by passive movement. How is this even possible?
- x 3. (a) Although all osmosis is diffusion, not all diffusion is osmosis. Argue. (b) Justify the fact that the cell theory briefly explains the concept of the cell and its applications.

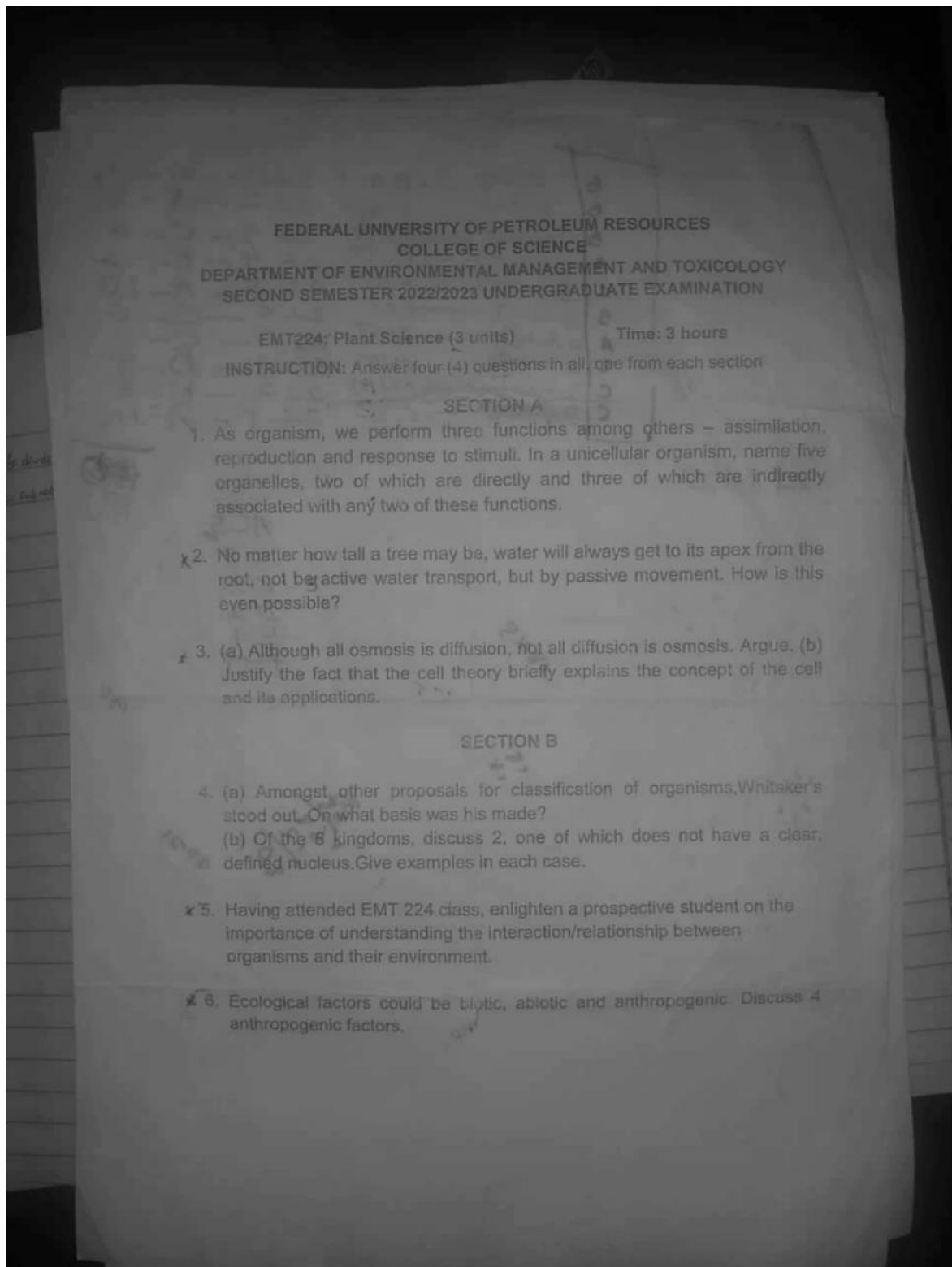
SECTION B

4. (a) Amongst other proposals for classification of organisms, Whitaker's stood out. On what basis was his made?
(b) Of the 5 kingdoms, discuss 2, one of which does not have a clear, defined nucleus. Give examples in each case.
- x 5. Having attended EMT 224 class, enlighten a prospective student on the importance of understanding the interaction/relationship between organisms and their environment.
- x 6. Ecological factors could be biotic, abiotic and anthropogenic. Discuss 4 anthropogenic factors.

Paper 3 of 5 — EMT 224 past question

Level 200 • Second Semester • PastQuestion

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Paper 4 of 5 — EMT 224

Level 200 • Second Semester • Handout

EMT224:

PRINCIPLE AND PRACTICE OF PHYTOREMEDIATION

Definition and Overview

Phytoremediation is the use of plants to remove pollutants from the environment.

Phytoremediation is a natural, cost-effective, and eco-friendly approach to cleaning polluted soil, water, and air. This innovative method leverages the power of living plants to absorb, break down, and remove a wide range of pollutants, including heavy metals, pesticides, industrial contaminants, and excess nutrients. By utilizing plants to clean pollutants, phytoremediation provides a sustainable solution that enhances ecosystem services, improves soil health, and supports biodiversity.

As a long-term solution, phytoremediation offers numerous benefits. It is a low-cost approach compared to traditional remediation methods, requiring minimal equipment, labor, and maintenance. Additionally, plants can be harvested and replaced, allowing for continuous pollutant removal and soil improvement over time. By harnessing the natural abilities of plants, phytoremediation provides an effective and environmentally friendly solution for pollution cleanup, making it an attractive option for revitalizing contaminated sites and promoting ecological balance.

Principles of Phytoremediation

Phytoremediation operates on the principle that plants can absorb, break down, and transform pollutants, making them harmless. Through their roots, stems, and leaves, plants take in pollutants from soil, water, and air, storing or degrading them. Harvesting and disposing of these plants removes pollutants from the environment. Different plant species are suited for specific pollutants and environments, allowing for targeted remediation.

The process is further enhanced by microorganisms in soil, which work in conjunction with plant roots to break down pollutants. Additionally, plant roots stabilize soil, preventing erosion and runoff, and promoting a healthy ecosystem. By leveraging these natural processes, phytoremediation provides an effective and sustainable solution for cleaning pollutants from the environment, restoring balance to contaminated sites.

Types of Phytoremediation

These include:

1. **Phytoabsorption:** Plants absorb pollutants through their roots, stems, and leaves.
2. **Phytoaccumulation:** Plants store pollutants in their tissues, reducing their availability in the environment.
3. **Phytodegradation:** Plants break down pollutants into less toxic compounds.
4. **Phytotransformation:** Plants convert pollutants into more stable, less toxic forms.
5. **Phytostabilization:** Plants stabilize pollutants in soil, preventing their migration.
6. **Phytovolatilization:** After contaminants have been degraded or modified in plant cells, plants release the pollutants into the air, where they can be dispersed or volatilized.

7. Rhizodegradation: Microorganisms in soil, associated with plant roots, break down pollutants.

Page 1 of 4

Applications of Phytoremediation

It can be applied in the following;

1. Soil Remediation: Cleaning polluted soil by removing heavy metals, pesticides, and industrial contaminants.
2. Groundwater Remediation: Removing pollutants from groundwater, making it safe for drinking and irrigation.
3. Wastewater Treatment: Using plants to treat and purify wastewater, reducing contaminants and pollutants.
4. Air Pollution Control: Absorbing and breaking down airborne pollutants, improving air quality.
5. Brownfield Restoration: Revitalizing contaminated industrial sites, making them safe for redevelopment.
6. Landfill Leachate Treatment: Treating toxic leachate from landfills, preventing groundwater pollution.
7. Agricultural Runoff Management: Reducing excess nutrients and pesticides in agricultural runoff, protecting waterways.
8. Urban Stormwater Management: Using plants to absorb and filter stormwater, reducing urban pollution.
9. Environmental Restoration: Restoring ecosystems damaged by pollution, promoting biodiversity and ecosystem services.
10. Climate Change Mitigation: Using plants to sequester carbon, reducing atmospheric CO2 levels.

Benefits of Phytoremediation

The following are some benefits of phytoremediation;

1. Improves air and water quality
2. Reduces soil pollution
3. Enhances ecosystem services
4. Increases biodiversity
5. Cost-effective cleanup solution
6. Low maintenance requirements
7. Permanent solution for pollution cleanup
8. Improves public health
9. Enhances community aesthetics
10. Supports sustainable land use
11. Can be used in conjunction with other remediation methods.

Plant Selection for Phytoremediation

Suppose one is called upon to make selection of particular plant species for use in

phytoremediation, there are some of the conditions for which these plants can be selected; these are characteristics the plants must have to fit into being used in the phytoremediation intervention;

1. High Biomass Production: The plant should be able to grow rapidly and produce a large amount of biomass to effectively absorb and remove pollutants.
2. Deep Root System: The plant should have a deep and extensive root network to absorb pollutants from deeper soil layers and prevent further contamination.
3. Tolerance to Pollutants: The plant should be able to withstand high levels of pollutants without suffering damage or dying off, ensuring continuous remediation.

Page 2 of 4

4. Hyperaccumulation: The plant should be able to accumulate high concentrations of pollutants in its tissues, making it an effective tool for removing contaminants.
5. Fast Growth Rate: The plant should grow rapidly to enable frequent harvesting and removal of pollutants, accelerating the remediation process.
6. Adaptability to Local Climate: The plant should be able to thrive in the local environmental conditions, including temperature, humidity, and soil type.
7. Drought Tolerance: The plant should be able to survive with minimal watering, reducing maintenance requirements and ensuring continued remediation during dry periods.
8. High Transpiration Rate: The plant should have a high water uptake rate to efficiently remove pollutants from the soil and groundwater.
9. Easy Propagation: The plant should be easy to propagate and reproduce, allowing for cost-effective and efficient large-scale cultivation.
10. Non-Invasive: The plant should not outcompete native species or spread beyond designated areas, preventing unintended ecological consequences.

Case Studies

I shall introduce 4 case studies from my research Team, including that with one of your lecturers here. Study at least 3 and see what was done and how it was done. Try to identify methodological flaws, or any form of critic, based on your experience;

1. Ikhajiagbe, B. (2016). Possible adaptive growth responses of *Chromolaena odorata* during heavy metal remediation. *Ife Journal of Science* 18(2): 406 – 411. (Published by Obafemi Awolowo University, Ile-Ife, Nigeria)
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Limitations and Challenges of Phytoremediation

The following are some limitations and challenges of phytoremediation;

1. Phytoremediation is a slow process that can take years to achieve significant remediation.
2. It is limited to surface contamination and may not be effective for deep-seated pollutants.
3. Plant growth and remediation rates can be affected by environmental factors like climate, soil type, and nutrient availability.
4. Some pollutants may be toxic to plants, limiting their use in phytoremediation.
5. Large-scale phytoremediation efforts can be costly and resource-intensive.
6. Public acceptance and regulatory frameworks may hinder the adoption of phytoremediation.

Page 3 of 4

7. Monitoring and maintenance requirements can be significant.
8. Phytoremediation may not completely eliminate pollutants, but rather reduce their concentrations.
9. Genetic modification of plants for enhanced remediation potential raises ethical concerns.
10. Integration with other remediation technologies may be necessary for comprehensive site cleanup.

All the best !

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Paper 5 of 5 — EMT 224

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